

PREETHAM REDDY AERRABOLU

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EDUCATION

Lawrence Technological University, Southfield, MI.

Master of Science, Mechanical Engineering.

Dec 2025

GPA-3.7/4.0

Courses:- Fatigue Analysis, MPC, Engineering Analysis, Advance Dynamic, Interaction design.

Osmania University, Hyderabad, TG.

BE in Mechanical Engineering.

June 2023

CGPA-7.12/10

Courses:- CAD/CAM/CAM, Mechatronics, Machine design & Material testing, Vehicle design, Production and operations management

ACADEMIC PROJECTS

Design and Analysis of Exhaust Cooling System for Heavy Military Vehicles.

April 2022 – March 2023

- Developed a highly effective cooling system for exhaust gases in military vehicles, which significantly improved engine performance and reduced overheating incidents and directly enhancing mission readiness and reliability.
- Directed a focused team in the development of an exhaust cooling system tailored for heavy military vehicles, incorporating advanced materials and engineering techniques using to effectively minimize thermal signatures during missions.
- Conducted rigorous testing and analysis, leading to a successful exhaust gas temperature reduction to 10°C; this achievement resulted in enhanced performance metrics and established a new benchmark for future projects.

SAESupra - Formula Vehicle Prototype

July 2021 – May 2022

- Contributed to the assembly and integration of critical systems in a prototype Formula vehicle, ensuring compliance with SAE regulations and standards; the vehicle was completed ahead of schedule, allowing for additional testing opportunities.
- Designed and analyzed the chassis subsystem, achieving a structural integrity rating exceeding 95% through rigorous testing protocols; optimized performance metrics led to a 15% improvement in handling during dynamic simulations.

RESEARCH EXPERIENCE

Research on Quadruped Robots – Master's Thesis.

May 2024 – Present

- Researched and analyzed the gaits of quadruped robots, focusing on hexapodal, bipedal, monopodal, and octopodal locomotion.
- Enhanced the walking capabilities of quadruped robots by modifying designs and simulating gait dynamics using **MATLAB, Sumolink, Ansys, Solidworks, 3D Printing**
- Developed leg design and stability control system for a quadrupedal robot (LRG), optimizing walking stability on varied terrain surfaces using Model Predictive Control (MPC) and calculating joint angles for precise locomotion.

WORK EXPERIENCE

LG Energy Solution

May 2025 – Present

- Supported EV battery pack design and testing, focusing on lightweight, non-conductive materials for end plates.
- Conducted BOM analysis and GD&T compliance for Ford C727 and Toyota RU battery systems.
- Designed and prototyped JIG fixtures (U611 lifting & pin removal), recognized for patent potential.
- Collaborated with Stellantis to update BOM-integrated drawings ensuring alignment with revised powertrain designs.
- Prepared structured documentation (CATIA V5 + Windchill) for onboarding, improving team integration.

Research Assistant -Dr. Hamid Vejdani – Lawrence Technological University Jan 2025 – Present

- Researched soft polymer actuators (HASEL/EH) for robotic legs using MATLAB, ANSYS, and 3D printing.
- Performed mechanical testing of soft actuators (tensile, compression, force-stroke) to evaluate energy-efficient locomotion.
- Authored an ASME IMECE 2025 paper: “Designing an Electrohydraulic Actuated Hopping Robot.”

Junior Mechanical Engineer R&D-Gear Head Motors, Hyderabad. Feb 2023 – Jan 2024

- Designed and tested universal EV battery box in SolidWorks/ANSYS ensuring compatibility across e-bikes.
- Conducted jump, compression, and safety testing on EV frames and battery packs; improved ergonomics based on user feedback.
- Led the development of an e-wheelchair attachment, reducing production cost from \$100 → \$30 through sheet metal optimization.

Sustainability Design Intern - Gear Head Motors, Hyderabad. August 2022 – January 2023

- Orchestrated the implementation of eco-friendly materials in disposable cutlery production, optimizing environmental impact by 35% and appealing to a broader, environmentally-conscious customer base, increasing sales by 18%.
- Conducted in-depth analysis of product's demand and supply requirements, resulting in a 15% decline in inventory costs and a 20% increase in stock turnover rate.
- Analyzed a comprehensive sustainability analysis, identifying 15 key areas for improvement; implemented changes that reduced waste and saved cost for the company.

Design Intern - MVSR Engineering College, Hyderabad. June 2020 – August 2020

- Created detailed 3D models and technical drawings for 20+ machinery parts using CATIA V5, SolidWorks and AutoCAD; improved production accuracy by 30% and cut down revision cycles by 15%.
- Spearheaded detailed analysis of 2-D figures and designed precise 3-D models, leading to an increase in design accuracy and reduction in prototyping costs.

EXTRA-CURRICULAR ACTIVITIES

- SAE (Society of Automotive Engineers) Student Coordinator.
- Participated in various tier events including Mobile Robotics, CAD Designing, Diagrammatical Reasoning.
- Member of National Service Scheme (NSS), a public service program in India.
- Team Member, College Basketball Team. Part of college motor sport team.

ACHIEVEMENTS

- Awarded Engineering Master’s Scholarship (\$3250/semester) at LTU.
- Recognized at LGESMI for patent-worthy JIG design in EV battery assembly.
- Active SAE Supra contributor; completed Formula vehicle ahead of schedule enabling extended testing.

SKILLS

ICT- AutoCAD, Fusion 360, SOLIDWORKS, Adobe Photoshop, C language, Python, CATIA V5, MATLAB, Ansys, MS Office, MS Excel, MS PowerPoint, Reverse Engineering, Robotics, Model Predictive Control (MPC), Ansys Workbench, APDL-Fluent, CAE, FEA, Autodesk Maya, Autodesk Alias, 3D Printing, Prototype Building.

CERTIFICATIONS - SOLIDWORKS | AutoCAD | Siemens NX | Fusion 360 | CATIA V5 | Ansys | MATLAB | 3D printing | Manufacturing Techniques.

LANGUAGES - Hindi (Proficient) | English (Proficient) | Telugu (Proficient).